Auteur: Anna Voronovska
Studierichting: Informatica
Oefeningenreeks 2D nr. 19.2

$$\lim_{x \to \pi} \frac{1 + \cos 2x}{3 \sin x} = \lim_{x \to \pi} \frac{1 + 2\cos^2 x - 1}{3 \sin x} =$$
$$= \lim_{x \to \pi} \frac{2\cos^2 x}{3 \sin x} = \frac{2\cos^2 \pi}{3 \sin \pi} = \frac{2 \cdot (-1)^2}{3 \cdot 0} = \frac{2}{0} = \infty$$

Wij kijken naar de linker en rechter limiet apart:

$$\lim_{x \to \pi^-} \frac{2\cos^2 x}{3 \sin x} = \frac{2}{0^+} = +\infty \qquad x < \pi \to \sin x \text{ positief}$$
$$\lim_{x \to \pi^+} \frac{2\cos^2 x}{3 \sin x} = \frac{2}{0^-} = -\infty \qquad x > \pi \to \sin x \text{ negatief}$$

References